A MODULAR LLM-BASED CONVERSATIONAL RECOMMENDER FOR EVENT MARKETPLACES

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Motivation

Event-marketplace users rarely begin with a neat filter query. They describe an event, a budget, a location, and multiple provider needs that arrive in any order.

- Manual filters expect users to know the catalog before they know the plan.
- LLM-only chat can produce fluent but unsupported provider claims.
- A production marketplace needs provider IDs, live evidence, contact actions, and traceability.

Research Question

Central question

What modular architecture lets natural-language event requirements become grounded provider recommendations in a real marketplace, and how reliable is it under live scenario-based evaluation?

Design thesis: the LLM interprets and writes; the typed runtime decides when to route, retrieve, persist, recommend, defer, or close.

Event-Plan-First State

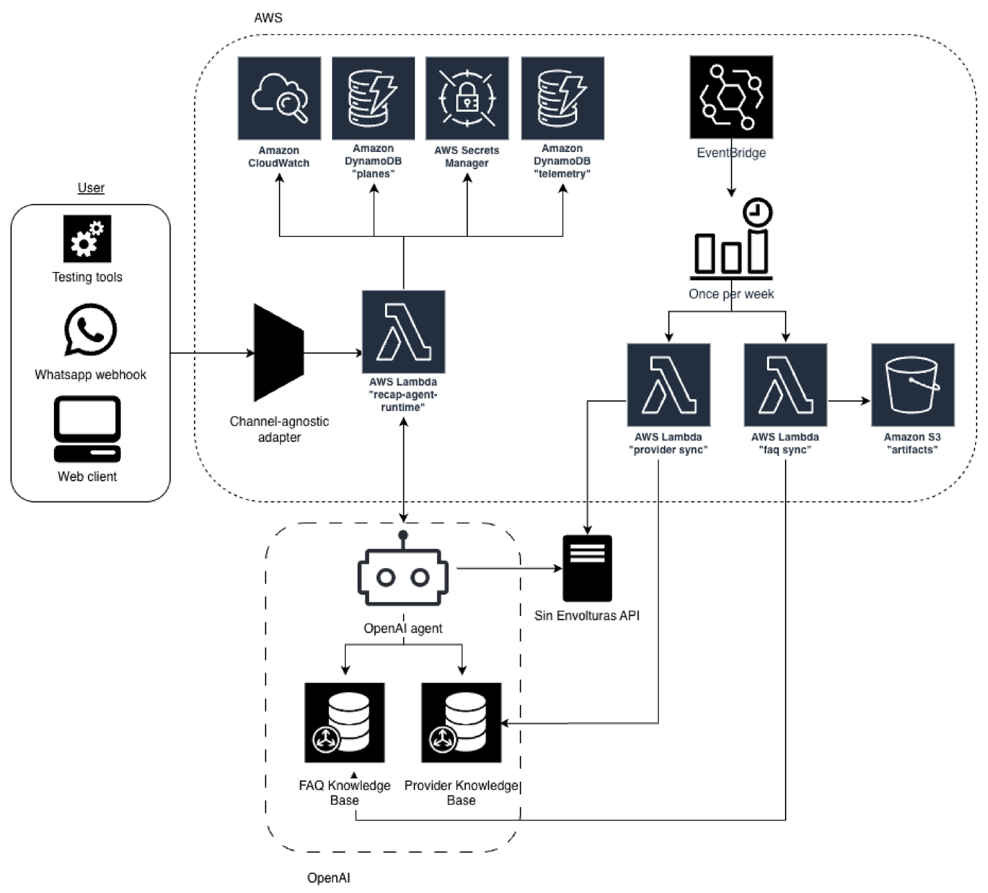
Plan fields	event type, date, location, guests, budget, preferences
Need fields	category, filters, shortlist, selected providers, status
Need states	identified, search-ready, shortlisted, selected, deferred, no results
Closure	contact validation, lead creation, partial closure support

The event plan is the durable artifact. It lets one conversation manage several provider needs without erasing earlier selections when a new need appears.

Technical Contributions

- Channel-agnostic runtime: WhatsApp behavior lives in adapters, not core flow logic.
- Typed workflow graph: route decisions are auditable state-machine outcomes.
- Grounded recommendations: provider text is tied to IDs and tool evidence.

System Architecture



Channel	WhatsApp-style adapters, webchat, terminal
Runtime	AWS Lambda, typed state, OpenAI Agents SDK
Evidence	Sin Envolturas APIs and vector stores
Storage	DynamoDB plans and telemetry

Runtime Pattern

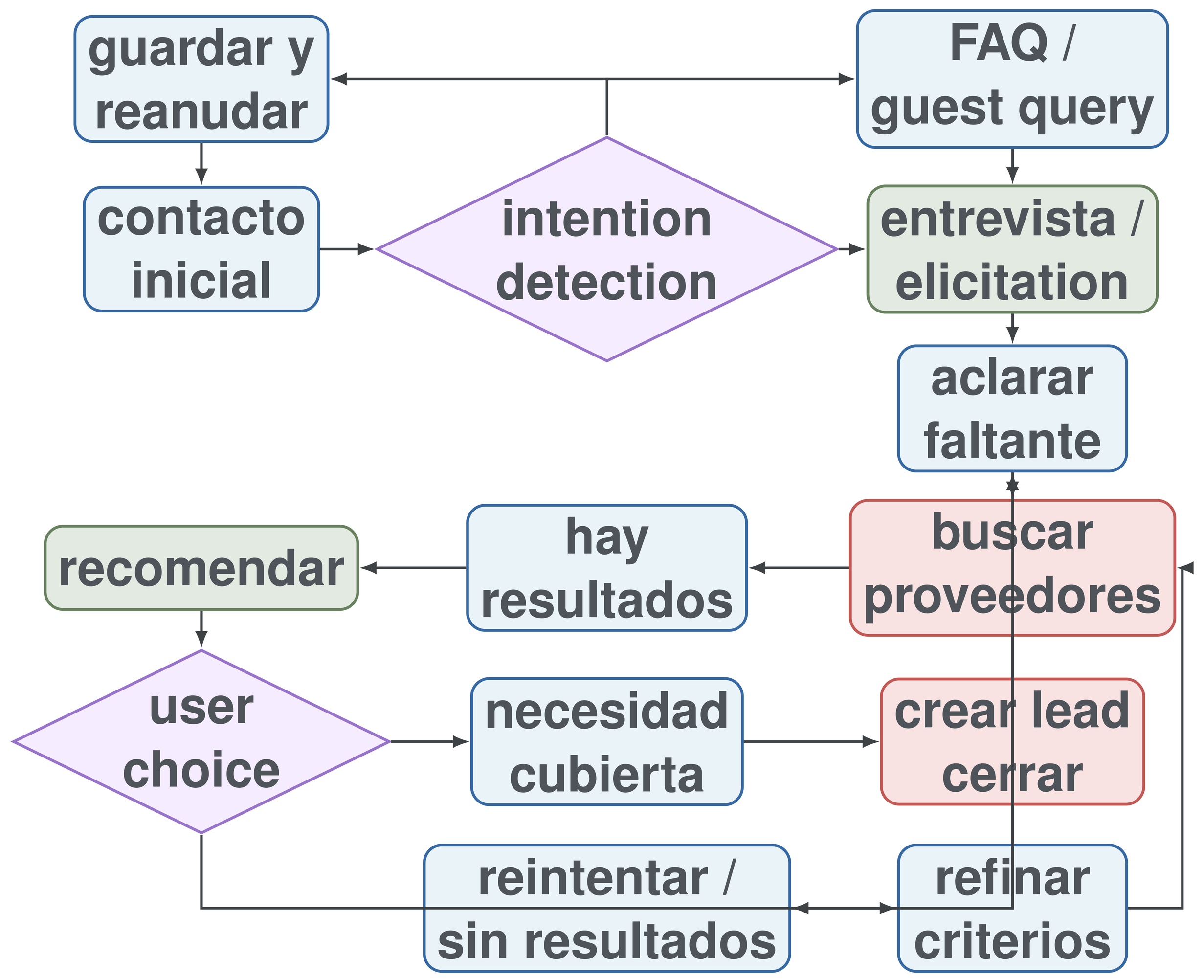
1. Load the persisted plan for the channel and external user.
2. Extract structured evidence from the Spanish user message.
3. Update the plan and compute a typed node decision.
4. Call only the tools allowed by that node and evidence.
5. Compose a bounded Spanish response and persist telemetry.

Flow decisions come from structured LLM extraction plus typed state-machine evidence. Deterministic code validates invariants and preserves already-established plan state.

Grounding Strategy

- Marketplace APIs remain the operational source of truth.
- Vector stores support semantic retrieval, not free-form claims.
- Recommendation turns require provider IDs and captured tool evidence.

State Diagram



The full implementation defines 26 decision nodes; this diagram compresses them into the main operational routes used by the live study.

Evaluation Design

Conversations	150 live Lambda runs
Scenarios	50 frozen Spanish scenarios, 3 repetitions
Event groups	wedding, birthday, baby shower, corporate, social
Route families	recommendation, clarification, multi-need, refinement, selection, pause/resume, closure, FAQ, no-results, recovery
Provider snapshot	182 real Sin Envolturas providers

Recommendation Funnel

Need extraction	159/162 = 98.15%
Recommendation after extraction	111/159 = 69.81%
End-to-end need coverage	111/162 = 68.52%

Headline Results

- **150** live conversations
- **88.67%** typed completion
- **0** runtime errors or timeouts
- **99.33%** persistence assertions
- **100%** category consistency
- **100%** grounding provenance

Route Reliability

Recommendation	15/15
Clarification	15/15
Multi-need planning	15/15
Refinement	15/15
Closure	15/15
Error recovery	12/15
Pause/resume	3/15

The Lambda was operationally stable; pause/resume behavior remains the clearest functional weakness.

Provider Evidence Checks

- **704/704** category match
- **0** known location mismatches
- **180/180** evidence-backed turns

Recommendations are constrained by provider IDs, tool outputs, and structured candidate provenance rather than generated provider descriptions alone.

Takeaway

Main message

Grounded conversational recommendation is not just prompt engineering. A reliable event-marketplace agent needs a bounded LLM, explicit workflow control, persistent state, live provider evidence, and audit-ready telemetry.

Next steps: enrich provider metadata, strengthen pause/resume, and compare against manual filters and forms.